Iris Phenolic Liners
Continuous Heating to Model Exhaustion

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How long until the gas reaches the casing?

For the sustained nominal load, the independent moving-wall calculation exposes the casing after about 104–111 s in the sustainer and 138–148 s in the booster. The clock starts when hot gas first reaches bare phenolic. Burnthrough occurs when the moving char surface reaches the casing.

Motor	Liner [mm]	Liner gone [s]	Earlier casing event
Sustainer Al	4.76	110.5	All Al liquid at 99.1 s
Sustainer CF	4.76	104.3	Casing reaches 1000 °C at 95.1 s
Booster Al	6.35	147.6	All Al liquid at 139.4 s
Booster CF	6.35	137.9	Casing reaches 1000 °C at 129.7 s

The calculation applies 2504 K gas continuously at 800 psi with an initial film coefficient of 1000 W/(m^2 K), gas blowing and zero hot-side radiation. The Al comparison follows pure-metal fusion and retains the molten metal in place. The CF comparison follows oxygen-free T700/M21 resin decomposition and holds the residue in place. The study has no measured Iris wall heat flux or fired-liner recession. All four times therefore describe the stated thermal scenarios. The casing can lose pressure capacity while phenolic remains.

What most affects the estimate. In the original backing comparison, doubling or halving the nominal film coefficient gives roughly 57–200 s for the sustainers and 75–266 s for the boosters. These endpoints belong to the baseline backing model. Changing to the more detailed backing at the nominal load shifts the estimates by 3.7–5.2 s for Al and about 0.3 s for CF. A firing heat-flux history and measured local recession would constrain the two-minute estimate much more directly.

Verification. The burnthrough intervals above come from a conservative radial finite-volume solver. Refinements and mass/enthalpy balances are given below. The independent coupon and three-dimensional input checks precede the native ANSYS comparison. The continuous motor event has yet to pass native verification.

1 Baseline comparison with fixed casing properties

With the original casing properties, the moving-wall calculation gives about 104–107 s for the sustainer and 138–142 s for the booster under the nominal continuous load. It advances the gas-facing radius until it reaches the casing. Char can remain after the virgin resin has decomposed, so conversion and casing exposure occur at different times. This is an independent finite-volume calculation. ANSYS verification of this coupled moving wall remains open.

Case	Initial liner [mm]	Nominal event [s]	Property flag [s]	Film scenarios [s]
Sustainer Al	4.7625	106.8	49.0	58.3–200.1
Sustainer CF	4.7625	103.9	20.5	56.5–194.9
Booster Al	6.3500	142.4	82.0	77.8–266.3
Booster CF	6.3500	137.7	35.1	74.7–258.3

Table 1: Nominal and film endpoints use 640 liner and 160 casing cells with a 0.02 s step ceiling. Film scenarios change the uncorrected coefficient from 2000 to 500 W/(m^2 K), respectively. The other inputs retain nominal values. The property flag in this table belongs to the nominal case.

All events occur after the assumed intact casing exceeds its provisional property range. The nominal gas temperature is 2504 K, nominal chamber pressure is 800 psi, and the nominal uncorrected film coefficient is 1000 W/(m^2 K). The calculation exposes the liner at ignition and sustains heating until full local loss. Zero hot-side radiation is assumed at nominal conditions. The outside sees a 22 °C environment, an 8 W/(m^2 K) film and emissivity 0.25. The extent of actual in-motor exposure is unknown.

The largest one-factor changes in this study come from wall heat transfer and the assumed energy cost of char removal. The times in the last column compare two specified heat-transfer scenarios. They do not quantify firing-to-firing variability or a safe firing interval. Firing heat flux versus time and recession measurements on the manufactured liner would directly constrain the two inputs driving this spread. Section 6 shows how accounting for aluminium melting changes the liner times and when the backing melts.

2 Remaining thickness and casing heat-up

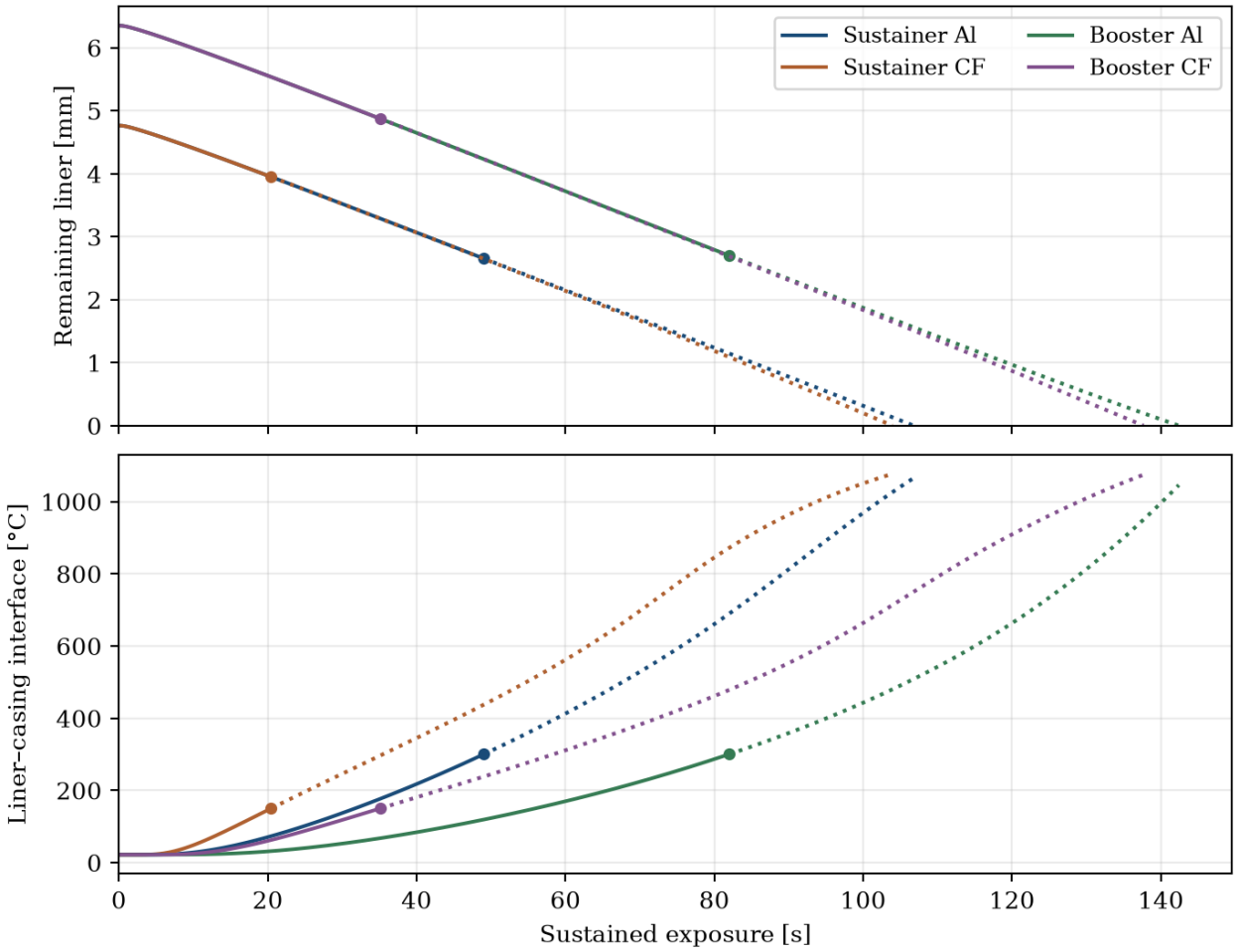


Figure 1: Moving-wall calculation at the finest nominal setting. Dots mark the first 150 °C CF or 300 °C Al property flag. Dotted lines continue with the inherited backing properties. The horizontal axis measures continuous exposure of the local wall.

The two casings give similar early surface recession because the heat initially penetrates only partway through the liner. The model separates them more clearly later as the casing absorbs heat. Each property flag denotes the limit adopted for using those casing data. It does not identify loss of pressure containment.

3 Temperatures, mass and energy in the sustainer Al calculation

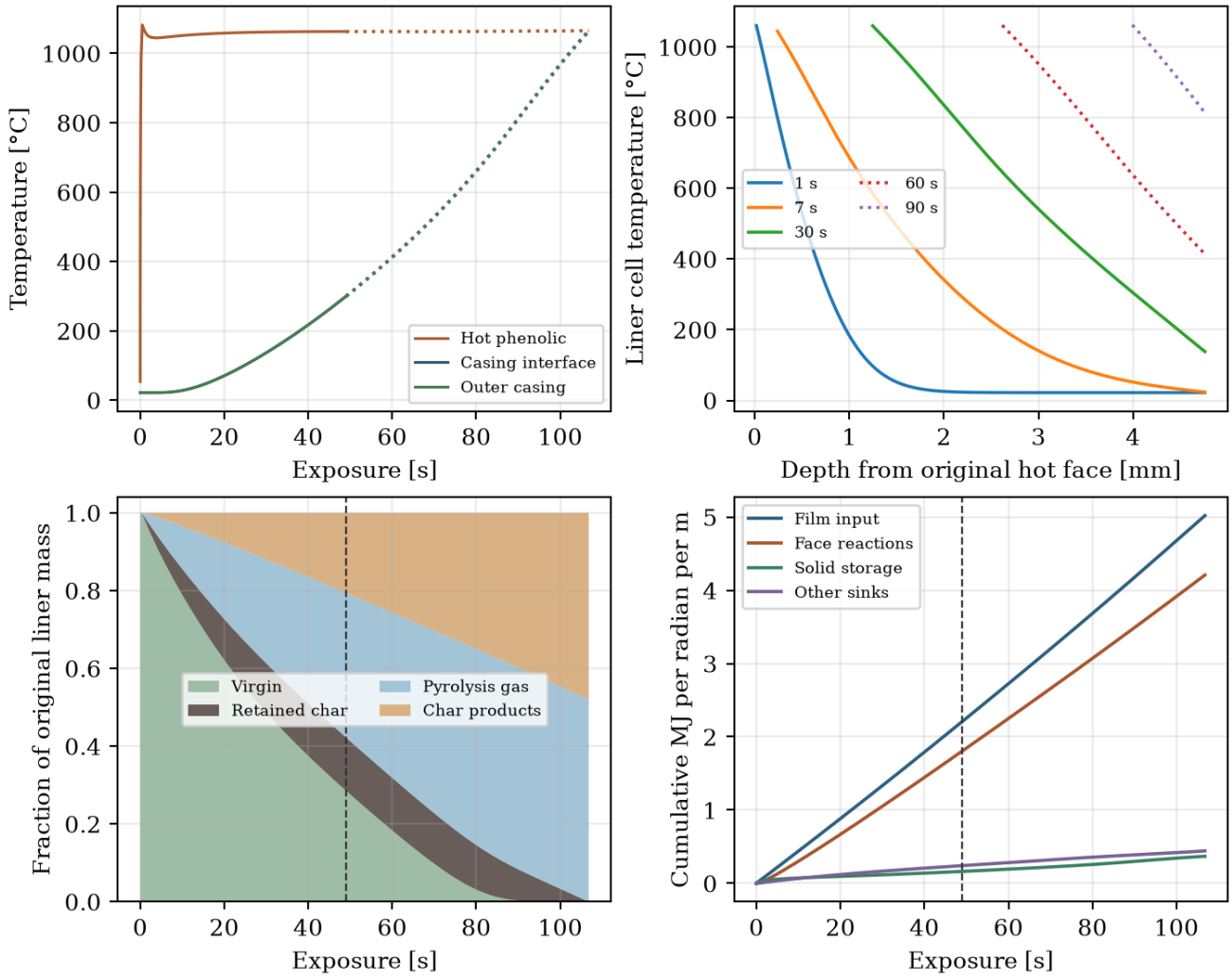


Figure 2: Upper panels show the three wall temperatures and the spatial temperature profiles of the surviving liner. Profiles start at the first cell centre. Lower panels account for the original liner mass and cumulative energy. Dashed vertical lines mark the casing property flag; subsequent values use surrogate continuation.

At 7 s the coupled wall gives 1048 °C at the hot phenolic face and 0.235 mm removed. At the same exposure time the accepted earlier native 320-row sustainer Al calculation gives 1444 °C at its live face and 0.045 mm of removed mesh. The gas is 2230.85 °C in both cases. The native calculation uses finite-row removal and its inherited reaction and energy equations. This difference therefore includes changes to the model as well as discretization; matching one final event time could conceal a substantial difference in surface behaviour.

At nominal exhaustion the surface reaction terms account for 83.8 percent of integrated film input in this case. The provisional 35 MJ/kg char cost therefore warrants a direct sensitivity check. Swept sensible heat appears on both sides of the interface accounting and cancels from the whole-wall ledger.

4 Numerical checks and open questions

Twenty nominal calculations isolate the time step at 160 cells, double the liner and casing radial counts twice at the same step, then halve the step on the 640/160 mesh. The last two comparisons below use 320 to 640 liner cells at 0.04 s and 0.04 to 0.02 s at 640 cells.

	Event shift [s]		Thickness	Mesh ΔT_s [K]	
Case	Mesh	Time	diff. [mm]	All samples	From 0.5 s
Sustainer Al	-0.0019	+0.0009	0.00016	30.3	2.4
Sustainer CF	-0.0018	-0.0003	0.00016	30.3	2.4
Booster Al	-0.0032	+0.0029	0.00024	39.3	4.7
Booster CF	-0.0030	+0.0000	0.00024	39.3	4.7

Table 2: Thickness difference is the largest across both comparisons at shared times. The temperature columns give the greatest hot-face difference between the two spatial resolutions, including or excluding the initial transient.

All eight finest pairs meet the 5 percent event-time target and the remaining-thickness target. Their final accepted time intervals overlap. Three coarser pairs have disjoint intervals. A load applied instantaneously to a massless reconstructed face gives a mesh-sensitive initial temperature. The 640-cell result still requires a separate comparison with ANSYS under matching equations, even where independent refinement differences are small.

The coupled implementation closes each cell energy equation before moving the front. The accepted step limits conversion growth to 0.05 and surface travel to 40 percent of the first-cell width while multiple cells remain. A remnant below half the original pitch merges conservatively with its neighbour. The last event is localized by solving for the step duration that brings the radius to the interface. A manufactured isothermal char front reproduces its analytical time to within 10^{-8} s at two resolutions. A nonuniform reacting coupon, interrupted after mesh mergers and a property flag, gives a byte-identical final state and history on restart. All twenty nominal integrations close the cumulative open-system energy and mass ledgers to relative residuals below 10^{-9} and 10^{-11}, respectively.

5 One-factor sensitivity to assumed inputs

Forty-two additional calculations each change one named input or one backing material tuple at 320/80 cells and a 0.04 s step ceiling. The columns show paired event times in the same order as their listed inputs, in seconds.

Case	Pressure 600 / 1000 psi	Radiation 0.1 / 0.3	Removal cost 17.5 / 70 MJ/kg	Gas excess 0.9 / 1.1
Sustainer Al	109 / 105	92 / 73	68 / 181	127 / 92
Sustainer CF	106 / 103	90 / 71	65 / 179	124 / 89
Booster Al	145 / 140	123 / 97	91 / 242	170 / 123
Booster CF	140 / 136	119 / 93	85 / 238	164 / 118

Radiation entries specify effective hot-side emissivity at the inherited gas radiative temperature. Gas excess scales the 2208.85 K difference between nominal gas and the initial wall, with convection coefficient and all other parameters fixed. The gas probes measure local sensitivity. Pressure enters the empirical consumption law; the film coefficient stays independent.

For each case, doubling and halving the film coefficient produce respectively the earliest and latest event in this screen. Both endpoints were rerun at 640/160 and 0.04 s, then 0.02 s (sixteen checks). The largest individual time shift is 0.007 s; 15 of sixteen pairs have overlapping final intervals. All sixteen satisfy the event-change and shared-thickness criteria. Individual endpoint intervals and the material crossings are recorded with the cases. An alternative inherited carbon-composite property tuple shifts the CF event by -0.84 s for the sustainer and -1.02 s for the booster. The resin, layup, bond and grade used on the actual cases remain unspecified.

Sections 6 and 7 examine aluminium fusion and CF resin decomposition with separate material surrogates.

6 Melting in a retained aluminium backing

The original aluminium cases use constant specific heat. The alternative stores the heat of fusion and allows the backing to become liquid while retaining its initial mass, shape, radial conductivity of 167 W/(m K) and density of 2700 kg/m^3. Gas loading, liner chemistry and dimensions match the nominal cases in Table 1. The extra heat absorbed by the casing delays liner exhaustion by 3.7 s in the sustainer and 5.2 s in the booster.

Case	Property flag [s]	First melt [s]	All cells liquid [s]	Liner gone [s]	Liner at full melt [mm]
Sustainer	50.7	84.1	99.1	110.5	0.43
Booster	84.4	124.4	139.4	147.6	0.23

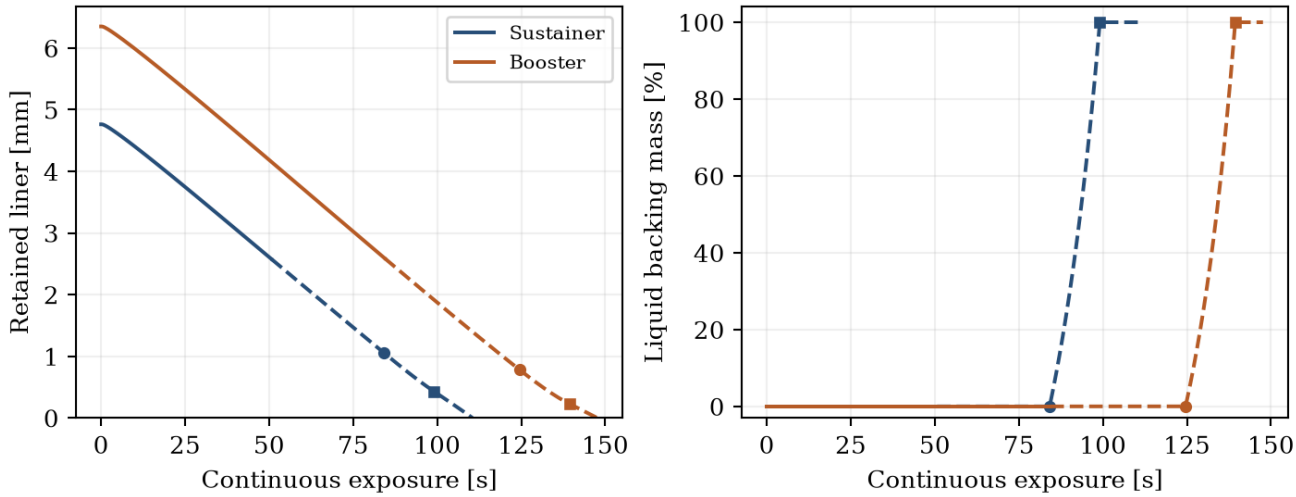


Figure 3: Solid curves end at the inherited 300 °C property flag. Dashes continue with the retained pure-Al surrogate. Circles mark the first cell with liquid and squares mark the first instant when all casing cells are liquid.

At complete modelled melting the liner still covers the case in both geometries. Following the liner further assumes the liquid stays in place. Pressure containment, flow of molten metal, alloy solidus and local wall defects need separate evidence.

The caloric law uses the crystal and liquid rows on the same NIST-JANAF enthalpy reference. At 933.45 K the 10.711 kJ/mol jump corresponds to 397.0 kJ/kg. The solver joins the two sensible branches across a narrow numerical melting interval, then recovers liquid fraction from the sharp equilibrium enthalpy. Ten integrations separate mesh, time-step and interval-width changes. The reported runs use 640/160 cells, a 0.02 s step ceiling and a 0.5 K interval. Across the eight single-parameter comparisons the greatest liner-event shift was 0.007 s. All eight satisfy the thickness and final-interval overlap checks. An externally heated char/Al coupon agrees with an independent sharp-transition enthalpy ODE to 0.25 K at the finest setting.

7 Resin reaction on the CF temperature histories

A separate calculation applies the two competing T700/M21 reactions measured under nitrogen to the liner-side and outside temperatures of the published CF cases. The heat histories are prescribed. Heat absorption, gas transport and the changing composite conductivity would have to feed back into the wall to revise a CF liner-exhaustion time.

Let $u = 1 - a_1 - a_2$ denote the fraction of the shared reactant still available. With T in kelvin and $k_j = A_j \exp[-E_j/(RT)]$ in s^{-1}, the rates are

$$\dot{a}_1 = k_1 u^{1.1125}, \qquad \dot{a}_2 = k_2 u^{2.0825}(1 + K_{\mathrm{cat}} a_2).$$

The frequency factors are 7.2193 and 5.3309×10^8 s^{-1}, and the activation energies are 58.7107 and 146.674 kJ/mol. The table uses $K_{\mathrm{cat}} = 7.4886$ from Table 5 of the experimental paper. The model equations and competitive mass-loss weights are given in its companion paper. Tranchard et al., Part I; Part II.

		Time to reactant conversion [s]			At 50%
Case	Casing location	10%	50%	90%	[°C]
Sustainer	Liner side	52.4	56.6	61.4	522
Sustainer	Outside	71.2	75.7	80.7	520
Booster	Liner side	80.5	85.9	91.8	514
Booster	Outside	117.5	123.2	129.8	511

Here 50% means half of the common reactive pool has advanced by either pathway. Reconstructing mass with the paper's density change and pathway weights gives about 12.3% loss of the initial composite mass at that point. The printed $\log_{10} K_{\mathrm{cat}} = -0.8744$ in Part I, Table 3 implies a smaller coefficient. Using that interpretation delays the four 50% times by 3.0–4.1 s. The calculations stop at 99.9% common conversion, 1000 °C, or the end of the supplied temperature history, whichever occurs first.

The local rates of temperature rise at 50% conversion are 492–697 °C/min, above the reported nitrogen tests at 5–100 °C/min. The Iris casing resin and layup are unspecified. These times identify when a reacting backing can affect the heat balance. The following section couples heat storage, radial conductivity, decomposition energy and escaping gas to both motor geometries.

8 CF casing heat, gas loss and liner exposure

The backing now feeds its changing mass, heat storage and through-thickness conductivity back into the thermal solution. Gas at 2504 K continues to heat the liner. The resin's two competing rates follow Section 7. Each casing cell releases gas at its own temperature into an assumed separate vent. This leaves the liner-side blowing calculation dependent on phenolic gas alone. The radial model holds the decomposed casing in its original shape.

Motor	Half reactive pool [s] Inner cell	Outer cell	Casing at 1000 °C [s]	Liner then [mm]	Liner gone [s]
Sustainer	60.7	82.4	95.1	0.45	104.3
Booster	91.4	135.3	129.7	0.40	137.9

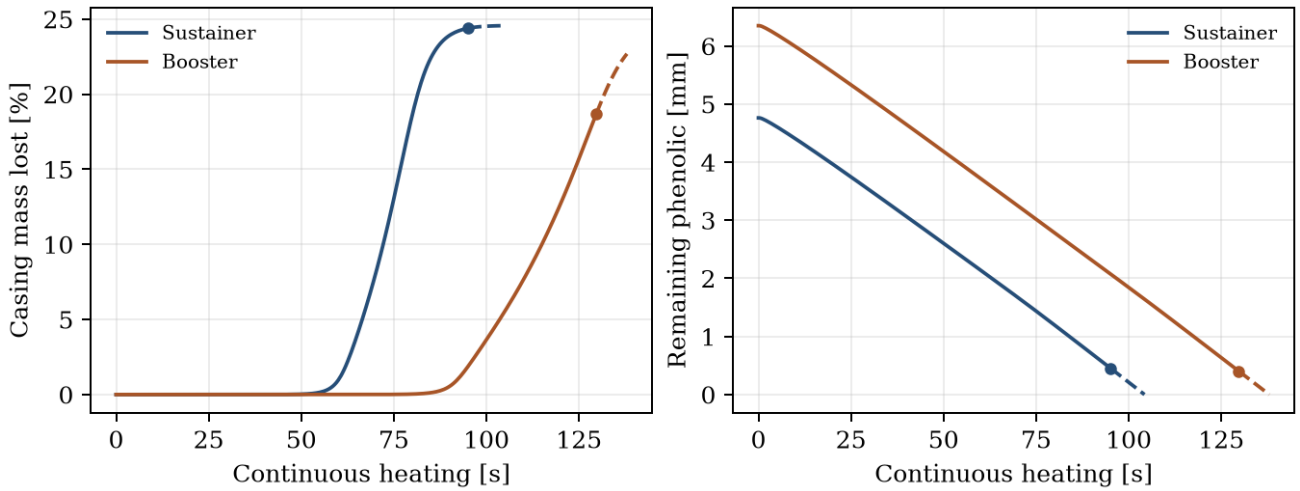


Figure 4: Gas release as a percentage of initial casing mass and remaining liner for the nominal reacting CF cases. Dots show the first backing temperature of 1000 °C. Dashed segments hold the upper-end residue and gas properties beyond this limit. Virgin-property fits reach their own limits at lower temperatures.

At liner exhaustion the surrogate releases 24.6% of the sustainer casing mass and 22.7% of the booster casing mass. The detailed backing changes the nominal liner time by +0.32 and +0.28 s respectively relative to the inert CF baseline. Changing the catalyst coefficient alone shifts these results by +0.084 and +0.122 s. The local resin, bond and layup in Iris are unknown. No structural failure time is inferred.

Local thermodynamics. For competing extents a_1, a_2, define $x = 0.554a_1 + 0.9479a_2$, $\rho = 1575 - 410x$ in kg/m^3 and $\Delta g_i = 410\chi_i \Delta a_i V$ in kg. The remaining cell mass is $m = \rho V$. With sensible enthalpies measured from 22 °C, the reaction substep closes

$$(mh_s)_{j+1} - (mh_s)_j = \sum_i Q_i \Delta g_i - h_g(T_{1/2}) \sum_i \Delta g_i.$$

Here $Q_1 = +259.54$ and $Q_2 = -152.22$ kJ per kg of generated gas, following the density-loss source in Tranchard et al., Part II. Positive Q_i adds heat. The model mixes virgin and degraded sensible enthalpies by x. Differentiating this stored energy includes its composition dependence.

9 CF transport properties and numerical checks

The model mixes normal conductivity and specific heat between virgin and degraded states using x. For temperature T in °C the virgin fits are $c_v = 687.31+2.8773T$ and $k_v = 0.61391+0.0011113T$. The corresponding upper fit limits are 300 and 150 °C. The frozen endpoint value is used outside either fit. The degraded and gas fits below use a 20–1000 °C interval, also with endpoint continuation:

$$
\begin{aligned}
c_d &= 662.53 + 2.599T - 2.0761\,10^{-3}T^2 + 5.1327\,10^{-7}T^3,\\
k_d &= 0.077211 + 2.3397\,10^{-4}T - 4.1903\,10^{-7}T^2 + 7.4228\,10^{-10}T^3,\\
c_g &= 1256.6 + 1.0610T - 9.2485\,10^{-4}T^2 + 3.5977\,10^{-7}T^3.
\end{aligned}
$$

Specific heats use J/(kg K) and conductivities W/(m K). Integrating each bounded heat-capacity function gives its sensible enthalpy. Local radial conductances include the logarithmic resistance of both adjacent annular half-cells. The thermal substep iterates conductivity with temperature. The adiabatic reaction substep advances both pathways against a common reactant pool and solves for temperature from the open-system energy equation. A trial repeats at a smaller time step if the reaction fails to converge or if any cell's total channel advance exceeds 0.04. The backing vent removes gas enthalpy once at generation. Gas pressure, internal transport work, swelling and delamination are outside this closure. Material measurements, Part I; thermal model, Part II.

Checks. A two-cell coupon with a constant outside heat input compares this split solver against an independent temperature-form ODE. Over 28 s the finest 0.01 s step gives at most 0.032 K temperature error and 0.0011 absolute error in either pathway. Both nominal motor pairs pass their mesh and time checks on the event brackets and remaining thickness. The finest comparisons use 640/160 liner/casing cells and a 0.02 s ceiling. Across the eight runs, each recorded profile independently reconstructs casing mass and total sensible enthalpy. An interrupted warm coupon recovered both reaction channels and produced identical final files after restart. These checks test the specified equations. Quantitative prediction for Iris still depends on a measured heat load, recession calibration and material-specific validation.

10 Comparison with the preceding finite-row model

For a continuously exposed liner, the inherited independent model reaches the casing after roughly two minutes for the sustainer and two and a half minutes for the booster. Each event occurs after the casing reaches the limit of its property evidence. Actual firing heat flux, local exposure duration and measured liner loss are unknown. The times below describe a hypothetical intact backing under sustained heating. Physical motor burnthrough remains open.

	Time to final liner-row loss [s]			Property flag [s]
Case	160, full step	160, half step	320, half step	320, half step
Sustainer Al	114.95	114.21	115.10	49.49
Sustainer CF	112.18	111.42	112.30	18.34
Booster Al	151.70	151.03	151.95	87.27
Booster CF	147.05	146.36	147.27	32.71

Table 3: Independent radial calculations. The initial thicknesses are 4.7625 mm for the sustainer and 6.3500 mm for the booster. A flag marks the first 150 °C CF or 300 °C Al interface crossing. The table separates this evidence limit from complete liner-row loss.

The gas stays at 2504 K (2230.85 °C), with nominal pressure 800 psi and uncorrected film coefficient 1000 W/(m^2 K). Outgassing modifies convection. Effective hot-side emissivity is zero. The outside sees 22 °C, a film coefficient of 8 W/(m^2 K) and emissivity 0.25. The hot surface temperature evolves as the wall absorbs heat. The local liner face is exposed from time zero until exhaustion. Sustained heating is a chosen scenario.

The three columns use 160/40, 160/40 and 320/80 radial cells in the liner and casing. The full step ceiling is 0.0015625 s and the half step is 0.00078125 s. In all twelve records the final row disappears within that run's time increment. Resolution changes move the event by more than an individual time increment, so digits beyond those shown carry little meaning.

11 Liner remaining and casing temperature

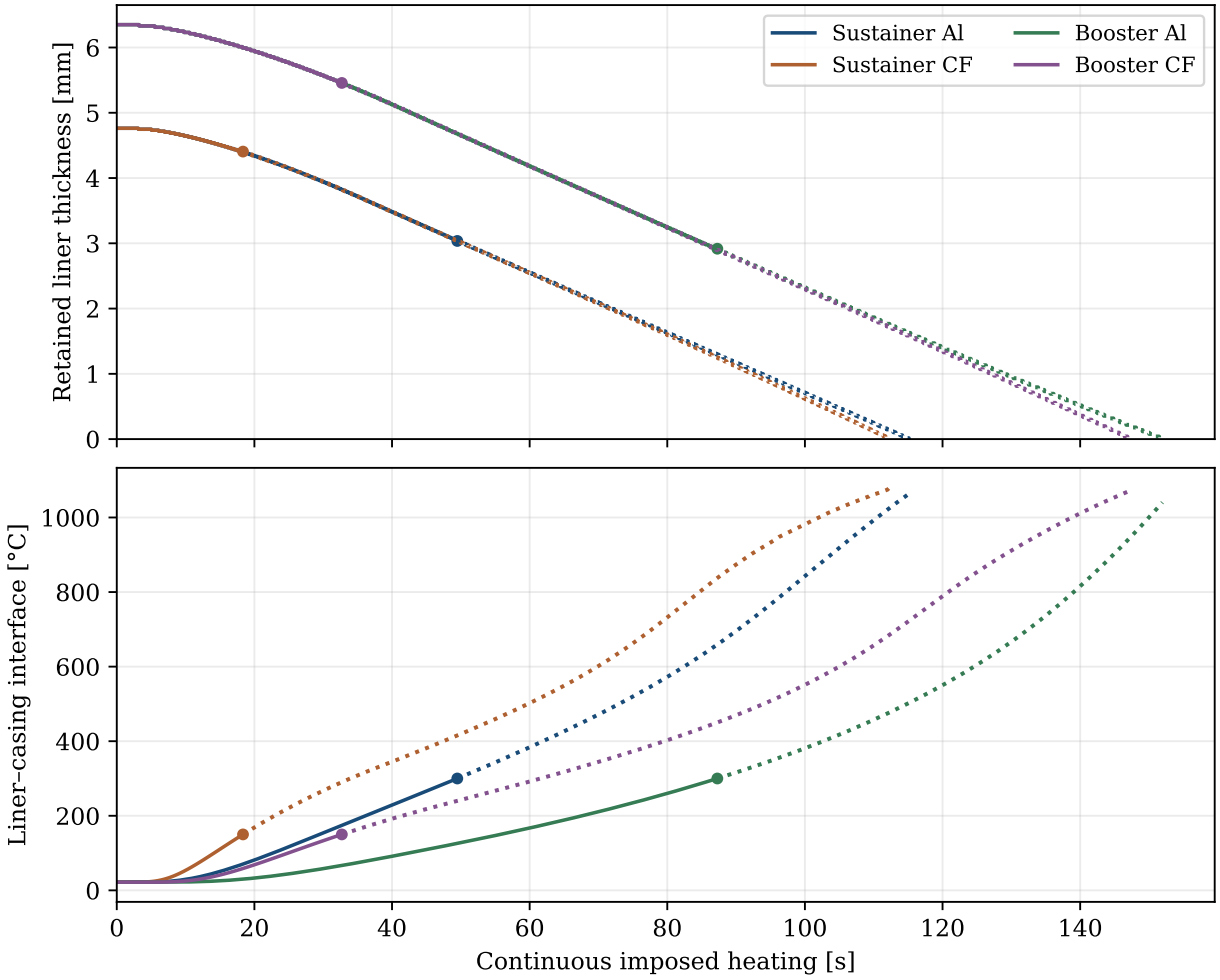


Figure 5: Independent 320/80 finite-row results. Dots mark the first casing-property flag. The dotted continuation assumes an intact casing with the inherited surrogate properties. Thickness includes retained char and is distinct from the thickness of virgin material. Similar curves overlap near ignition.

Al uses the inherited constant properties. The CF backing uses its radial conductivity and a heat-capacity correlation clamped to the inherited interval. The plotted temperatures can exceed the materials' supported range by hundreds of degrees. Integrity of a hot pressurized case, resin decomposition, bond failure and local cracks require additional evidence. The previous 7 s pulse and cooling results, native ANSYS verification and complete MAPDL and Fortran listings are in the companion native report. Earlier rejected endurance jobs are excluded from every comparison here.

12 What changes with time step and mesh?

Time comparisons halve the time step at fixed 160/40 mesh. Mesh comparisons double the number of cells in both materials at the same half step. All other physics inputs match between each pair. A signed time change is the finer result minus the coarser result.

	Event change [s]		Remaining thickness [mm]		
Case	Time	Mesh	Time diff.	Mesh diff.	Target
Sustainer Al	-0.74	+0.89	0.0595	0.0595	0.0476
Sustainer CF	-0.76	+0.88	0.0595	0.0595	0.0476
Booster Al	-0.67	+0.93	0.0397	0.0595	0.0635
Booster CF	-0.69	+0.91	0.0397	0.0595	0.0635

Table 4: Thickness columns give the largest absolute difference on a shared 0.5 s grid up to the first exhaustion. Every row transition was archived, so sampling uses the retained row count at each comparison time.

The largest change in event time is 0.78 percent of the finer calculation. 4 of eight comparisons meet the intermediate-thickness target and 0 of eight pairs have overlapping final-event brackets. Both sustainer pairs miss the 0.047625 mm target. Both booster pairs meet their 0.0635 mm target on the shared grid and over the full common history. Halving the step advances the event in all four cases, while doubling the mesh count delays it. Comparing only the first and third columns would partially cancel those effects. Each row retains its full thermal capacity during partial consumption until its residual-mass criterion removes it. The moving-wall treatment instead tracks the face continuously.

A separate thin-coupon test paused the independent worker after the second row loss and casing flag. A fresh controller then recovered the same complete history, final-event bracket and byte-identical terminal checkpoint as uninterrupted integration. The audit reconstructs final mass from the saved states, checks each row transition, and matches source and input fingerprints across all twelve records. Maximum cumulative relative mass error is 1.54e-14. Maximum relative residual of the inherited discrete step energy equation is 9.22e-11. The new material and wall equations below use a separate open-system energy ledger. A new 585-node native conduction verification input has been sealed for ten-core ANSYS execution after the active job. The independent coupled checks appear in the opening sections. Native comparison under matching moving-wall equations is still pending. Measured heating and recession, high-temperature backing data and sensitivity to surface chemistry remain necessary for the physical interpretation.

13 A moving surface with separate virgin, char and gas inventories

The candidate transports the masses of virgin phenolic M_v, char M_c, and their combined sensible enthalpy H_s in each cylindrical cell. A cell has volume $(r_o^2 - r_i^2)/2$ per radian per metre of axial length. Conservative overlap integration in $r^2/2$ returns both the retained and swept inventories as the face advances. In the equations below, M and H may refer either to a physical cell or to these quantities per radian per metre, provided every source uses the same convention.

13.1 Bulk reaction and reconstruction

We use a common reference $T_* = 22\,^\circ\mathrm{C}$ for virgin solid, char and gas. The inherited table defines piecewise-linear specific heat capacities. Let $h_i(T) = \int_{T_*}^{T} c_{p,i}(u)\,du$ for $i = v, c$ and $h_g(T) = 1600(T - T_*)$ J/kg when T is in degrees Celsius. The candidate holds the first heat capacity constant between 0 and 22 degrees Celsius and rejects states above the final 2476.85 degree table entry or below zero. Positive phase masses give a strictly increasing enthalpy relation $H_s = M_v h_v(T) + M_c h_c(T)$. Inverting the quadratic integral within its table interval recovers a single temperature after remapping. An exhausted cell carries zero mass and enthalpy and has no temperature to reconstruct.

Let $y = 600/1250 = 0.48$ denote the assumed char yield per kilogram of virgin material consumed. With $\bar{T} = (T^n + T^{n+1})/2$ and $k = 333\exp[-64081/(R(\bar{T} + 273.15))]$ in reciprocal seconds, the update uses

$$d = M_v^n(1 - e^{-k\Delta t}), \quad M_v^{n+1} = M_v^n - d, \quad M_c^{n+1} = M_c^n + yd, \quad m_g = (1 - y)d.$$

Mass and enthalpy use the same d. The local scalar temperature solve, and the converged coupled cell equation, enforce

$$H_s^{n+1} - H_s^n = Q_{\mathrm{cell}} - L_p d - m_g h_g(\bar{T}).$$

The chosen basis for $L_p = 418$ kJ/kg is virgin mass consumed at T_*. Gas leaves carrying its sensible enthalpy at generation. The phase inventories on the left already include the change in stored solid sensible heat.

13.2 Partially converted material at the wall

Let a and b be the virgin and char mass densities in the material adjacent to the gas. The current candidate assumes that residual virgin material swept by the front completes pyrolysis there and its char reacts at that same face. For speed $v \geq 0$ the outgoing surface pyrolysis gas flux is $j_g = (1 - y)av$ and the outgoing char-product flux is $j_c = (b + ya)v$, in kg/(m^2 s). Their sum equals the rate $(a + b)v$ at which the surface sweeps up solid. Writing $\alpha = b/(b + ya)$, we impose $j_c = \alpha j_{\mathrm{law}}(T_s, p)$ with the inherited kinetic and mass-transfer law. A virgin wall can heat and pyrolyse volumetrically before surface recession begins.

For a first-cell temperature T_1, define the sensible power delivered to the boundary by swept inventory as $q_{\mathrm{swept}} = v[ah_v(T_1) + bh_c(T_1)]$. Per unit hot-face area the wall balance is

$$q_{\mathrm{conv}} + q_{\mathrm{rad}} + q_{\mathrm{swept}} = q_{\mathrm{cond}} + j_g h_g(T_s) + j_c h_c(T_s) + avL_p + j_c L_c,$$

$$q_{\mathrm{cond}} = \frac{\displaystyle\int_{T_1}^{T_s} [(1 - \alpha)k_v(T) + \alpha k_c(T)]\,dT}{r_s \log(r_1/r_s)}.$$

Here $q_{\mathrm{cond}} > 0$ flows into the retained solid, r_s is the moving wall radius and r_1 the adjacent cell centre. The conduction expression integrates the cylindrical resistance at fixed local composition. The film coefficient uses total outward mass flux, including bulk gas arriving at the surface and both j_g and j_c. A bracketed scalar solve closes film heating, optional radiation, conduction and recession at the same T_s. Bulk gas has already exported its sensible heat in the bulk equation. The chosen proxy $h_c(T_s)$ for the char product and excess cost $L_c = 35$ MJ/kg together specify a provisional removal energy. Data for reaction products, recession and pyrolysis are needed to determine whether these assumptions hold for the Iris liner.

13.3 Verification to date

Tests compare the enthalpy primitive against independent quadrature and the isothermal reaction against the analytic solution. In a four-second manufactured heated cell, doubling the time steps from 8 to 16, 32 and 64 reduces temperature error against an independent ODE integration from 0.151 to 0.0376, 0.00940 and 0.00235 K. A nonuniform reacting shell with prescribed recession closes the integrated mass and energy ledgers through final-cell removal to relative errors below 2×10^{-15}. Separate manufactured tests recover imposed face temperatures for a virgin conducting wall, pure reacting char, and a mixed wall with blowing and radiation. The face equation residual is below 3×10^{-9} W/m^2 for those fixtures. These checks exercise individual material and boundary balances. The opening sections give the coupled transient checks and event comparisons. Native verification of this formulation remains open.

13.4 Radial heat flow between cells

On a stationary mesh the conduction operator transfers heat between adjacent cell centres through the two cylindrical half-cell resistances. For centre radii r_i, r_{i+1} and shared face r_f, the heat flow per radian per axial metre toward increasing radius is

$$F_{i+1/2} = \frac{T_i - T_{i+1}}{\log(r_f/r_i)/k_i + \log(r_{i+1}/r_f)/k_{i+1}}.$$

Backward Euler balances each cell's change in sensible heat against the difference of its two face flows and any supplied power. A two-layer steady wall reproduces the logarithmic temperature solution to within 6×10^{-13} K. A transient annular conduction test compares against an analytic Bessel eigenmode over one decay time. With the time error removed to first order, the maximum errors on 16, 32 and 64 cells are 0.118, 0.0295 and 0.00738 K. These tests exercise diffusion and the material interface at fixed geometry. The coupled checks extend these tests to simultaneously evolving temperature, gas flow and surface position.

Reference for the method. Chen and Milos describe a moving-grid treatment of decomposition, energy and surface balances for a charring ablator. Their PICA data apply to a different material and environment. NASA NTRS 20130014432.

A Conservative geometry and remapping code

Below is the complete independent geometry kernel. The accompanying model directory contains the bulk reaction, boundary and conduction implementations and their verification programs.

```
"""Conservative radial geometry kernel for a future moving-surface model.

This is NOT a heat-transfer/reaction solver or a recession law. It transports
extensive states under a prescribed boundary movement. Units are per radian
and per metre of axial length; multiply by sector angle and length if needed.
"""
from dataclasses import dataclass
import numpy as np


@dataclass
class Remap:
    density: np.ndarray
    cell_integrals: np.ndarray
    removed_integrals: np.ndarray
    original_integrals: np.ndarray
    balance_residual: np.ndarray


def advance_front(radius, interface, speed, dt):
    """Locate geometric exhaustion for a prescribed constant normal speed.

    The caller must obtain speed from a verified surface mass/energy balance.
    A temperature, conversion threshold or arbitrary cutoff is not used here.
    """
    if not np.all(np.isfinite([radius, interface, speed, dt])):
        raise ValueError('Nonfinite front inputs')
    if radius <= 0 or interface < radius or speed < 0 or dt <= 0:
        raise ValueError('Invalid front inputs')
    if radius == interface:
        return interface, 0., True
    if speed == 0:
        return radius, dt, False
    to_event = (interface - radius) / speed
    if to_event <= dt:
        return interface, to_event, True
    return radius + speed * dt, dt, False


def remap(old_edges, density, new_edges):
    """Integrate piecewise-constant conserved densities over new radial cells.

    The last dimension of density is the old cell index. Other dimensions can
    hold virgin-equivalent mass, char mass, solid enthalpy and related conserved
    quantities. Remap those quantities, not temperature or alpha independently.
    Thermodynamic reconstruction and reaction/removal source terms are the
    caller's responsibility. Interface positions must be retained by the caller
    so heterogeneous material labels are not averaged into one material.

    New cells cover the retained domain [new inner radius, old outer radius].
    A one-entry new_edges array denotes exact exhaustion of this domain. The
    swept-out mass and enthalpy are returned explicitly, not silently deleted.
    """
    old = np.asarray(old_edges, dtype=float)
    new = np.asarray(new_edges, dtype=float)
    field = np.asarray(density, dtype=float)
    if old.ndim != 1 or new.ndim != 1 or len(old) < 2 or len(new) < 1:
        raise ValueError('Expected radial edge arrays')
    if field.ndim < 1 or field.shape[-1] != len(old) - 1:
        raise ValueError('Density field does not match old cells')
    if not all(np.all(np.isfinite(x)) for x in [old, new, field]):
        raise ValueError('Nonfinite remap inputs')
    if old[0] <= 0 or np.any(np.diff(old) <= 0) or np.any(np.diff(new) <= 0):
        raise ValueError('Cells must have strictly positive radial volume')
```

```
    if new[0] < old[0] or new[-1] != old[-1]:
        raise ValueError('Only␣inner␣recession␣with␣a␣fixed␣outer␣boundary␣is␣supported')
    # x = r^2/2 is the exact cylindrical volume coordinate per radian/metre.
    xold = old * old / 2
    xnew = new * new / 2
    old_integrals = field * np.diff(xold)
    cumulative = np.concatenate([np.zeros(field.shape[:-1] + (1,)),
                                 np.cumsum(old_integrals, axis=-1)], axis=-1)
    flat = cumulative.reshape((-1, len(old)))
    retained_cumulative = np.array([np.interp(xnew, xold, row) for row in flat])
    retained_cumulative = retained_cumulative.reshape(field.shape[:-1] + (len(new),))
    integrals = np.diff(retained_cumulative, axis=-1)
    removed = retained_cumulative[..., 0]
    original = cumulative[..., -1]
    residual = original - removed - integrals.sum(axis=-1)
    new_density = integrals / np.diff(xnew)
    return Remap(new_density, integrals, removed, original, residual)
```